Abstract

The Traffic Signal Countdown System is a digital system designed to control traffic signals at a two-way road intersection while displaying the remaining time of the current signal. The system controls traffic in two directions: North-South (NS) and East-West (EW).

The traffic controller consists of four states: NS Green, NS Yellow, EW Green, and EW Yellow. The system changes from one state to the next automatically after the required time has passed. The Green signal has a duration of 10 seconds, while the Yellow signal has a duration of 4 seconds. A countdown display is used to show the remaining time, with Green counting from 9 to 0 and Yellow counting from 3 to 0.

The system was designed using Verilog HDL and tested through functional simulation. A corresponding circuit was also implemented in Proteus using digital logic components. In the Proteus implementation, a counter is loaded with the required starting value and counts down with each clock pulse. When the counter reaches zero, the traffic signal changes to the next state and the counter is loaded with the next required value.

The project demonstrates the practical use of combinational logic, sequential logic, counters, Finite State Machines (FSMs), and 7-segment displays in a real-world digital system.

Introduction

Traffic signals are used at road intersections to control the movement of vehicles and prevent conflicting traffic movements. A traffic signal controller needs to follow a specific sequence so that traffic from different directions is given permission to move at the correct time.

In this project, a Traffic Signal Countdown System was designed for two directions, North-South and East-West. The system uses a fixed sequence of traffic-light states. When one direction has a Green signal, the other direction has a Red signal. Before changing to the other direction, the Green signal changes to Yellow.

The system has four main states:

1. North-South Green
2. North-South Yellow
3. East-West Green
4. East-West Yellow

The sequence continuously repeats:

NS Green→NS Yellow→EW Green→EW Yellow

A countdown mechanism is also included to show how much time is remaining before the current state changes. For the Green state, the countdown displays values from 9 to 0, while the Yellow state displays 3 to 0.

The project was implemented in two ways. First, the traffic signal controller was designed using Verilog HDL and its operation was verified through simulation. Second, a corresponding digital circuit was created in Proteus using hardware components such as a clock generator, counter, comparator, 7-segment display driver, and LEDs.

This project therefore provides a practical application of the concepts studied in Logic and Sequential Circuit Design.

Problem Statement

At a road intersection, traffic from two different directions cannot be allowed to move simultaneously. A system is therefore required to control the traffic lights in a fixed sequence and ensure that the correct signal is activated for each direction.

The problem addressed in this project is to design a digital traffic signal system that can:

- Control traffic lights for the North-South and East-West directions.
- Provide Red, Yellow, and Green signals for both directions.
- Automatically change between the different traffic-light states.
- Keep each state active for a specific amount of time.
- Display the remaining time using a 7-segment display.
- Ensure that conflicting directions do not receive Green signals at the same time.
- Continuously repeat the traffic signal sequence.
- Be tested through Verilog HDL simulation.
- Be implemented and verified as a digital circuit in Proteus.

The system should therefore combine sequential control, timing, counting, and combinational output logic to produce a complete working traffic signal controller.

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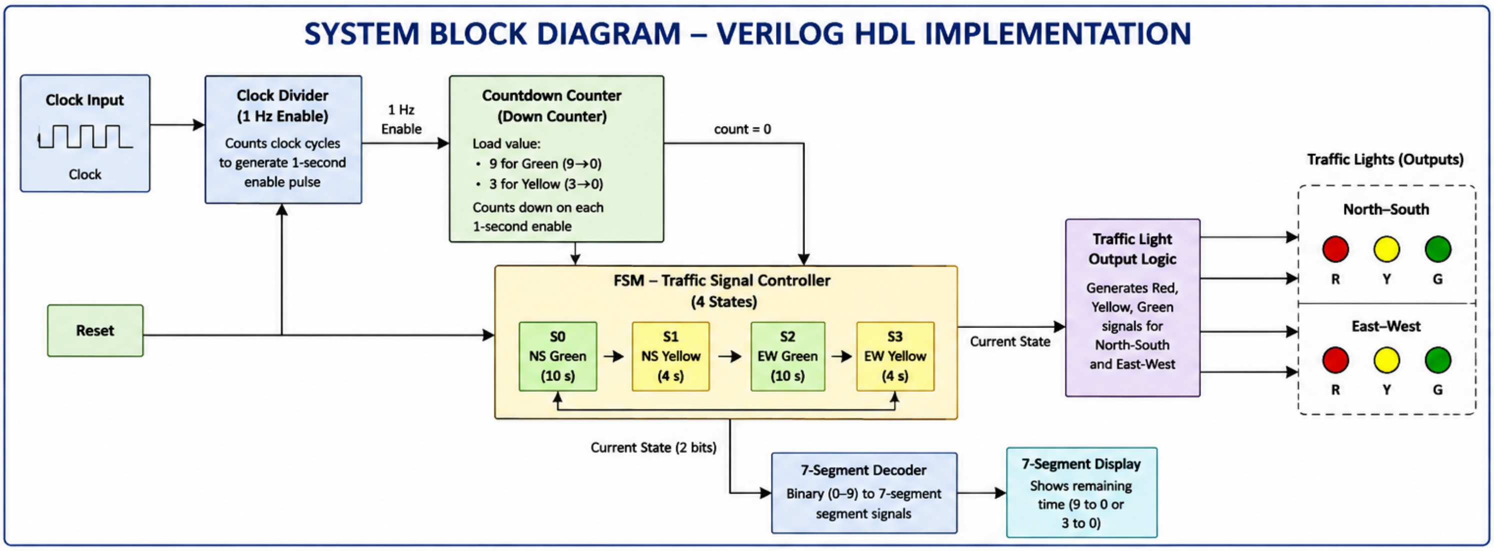
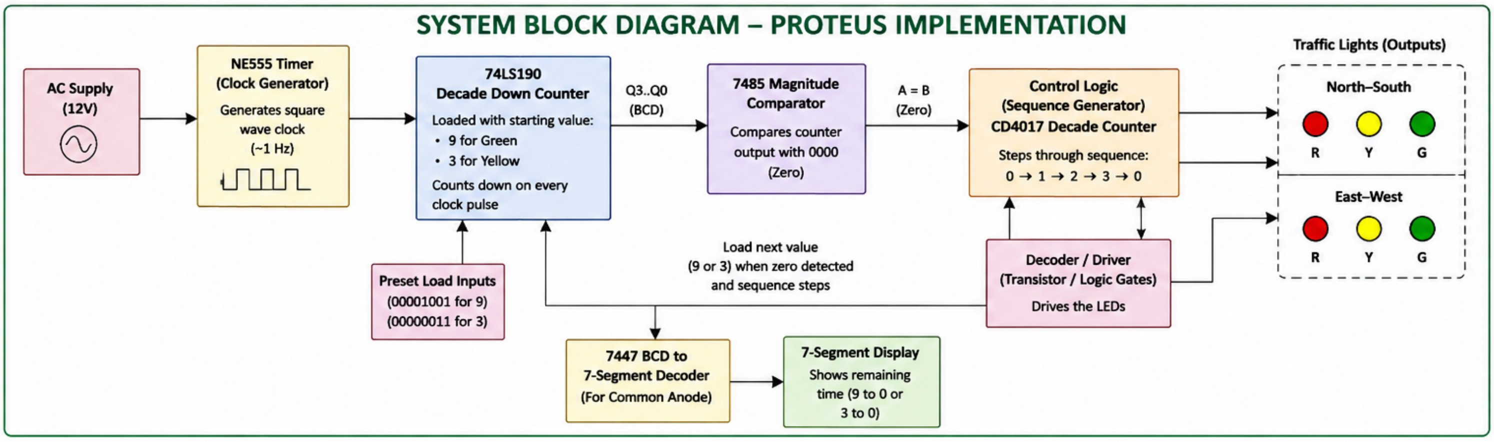
Objectives

The main objectives of this project are:

1. To design a digital traffic signal controller for a two-direction road intersection.
2. To implement a four-state Finite State Machine (FSM) for controlling the traffic signal sequence.
3. To control the Red, Yellow, and Green signals for both the North-South and East-West directions.
4. To implement a countdown system that keeps track of the remaining time for each traffic-light state.
5. To display the countdown on a 7-segment display so that the remaining signal time can be easily observed.
6. To use sequential logic and counters to control the timing and state transitions of the system.
7. To derive and simplify the required Boolean expressions for the traffic-light outputs using truth tables and Karnaugh maps.
8. To implement the system using Verilog HDL and verify its operation through functional simulation.
9. To implement a corresponding circuit in Proteus using digital logic components.
10. To compare and verify the operation of the simulated and circuit implementations and ensure that the traffic signals follow the required sequence correctly.

System Block Diagram





Truth Tables and Boolean Expressions

The Traffic Signal Countdown System uses four states to control the traffic lights. The states are:

- NS_GREEN: North-South traffic has a Green signal.
- NS_YELLOW: North-South traffic has a Yellow signal.
- EW_GREEN: East-West traffic has a Green signal.
- EW_YELLOW: East-West traffic has a Yellow signal.

The traffic signal sequence is:

NS_GREEN→NS_YELLOW→EW_GREEN→EW_YELLOW→NS_GREEN
Here, 1 represents ON and 0 represents OFF.

Traffic Light Truth Table

Current State	NS Red	NS Yellow	NS Green	EW Red	EW Yellow	EW Green
NS_GREEN	0	0	1	1	0	0
NS_YELLOW	0	1	0	1	0	0
EW_GREEN	1	0	0	0	0	1
EW_YELLOW	1	0	0	0	1	0

The table shows that only the required traffic signal is turned ON for each state. When one direction has a Green or Yellow signal, the other direction has a Red signal. This prevents both directions from receiving a Green signal at the same time.

State Transition Table

The FSM changes to the next state when the countdown reaches zero.

Current State	Condition	Next State
NS_GREEN	Countdown = 0	NS_YELLOW
NS_YELLOW	Countdown = 0	EW_GREEN
EW_GREEN	Countdown = 0	EW_YELLOW
EW_YELLOW	Countdown = 0	NS_GREEN

Therefore, the system continuously follows the sequence:

NS_GREEN→NS_YELLOW→EW_GREEN→EW_YELLOW→NS_GREEN

Boolean Expressions

The Boolean expressions for the traffic-light outputs can be obtained from the traffic-light truth table. Each output is active only during its corresponding state.

North-South Green
The North-South Green light is ON only when the system is in the **NS_GREEN** state.
NS_Green = NS_GREEN

North-South Yellow
The North-South Yellow light is ON only when the system is in the **NS_YELLOW** state.
NS_Yellow = NS_YELLOW

North-South Red
The North-South Red light is ON during both East-West states.
NS_Red = EW_GREEN + EW_YELLOW

East-West Green
The East-West Green light is ON only when the system is in the **EW_GREEN** state.
EW_Green = EW_GREEN

East-West Yellow
The East-West Yellow light is ON only when the system is in the **EW_YELLOW** state.
EW_Yellow = EW_YELLOW

East-West Red
The East-West Red light is ON during both North-South states.
EW_Red = NS_GREEN + NS_YELLOW

Output	Boolean Expression
NS Green	(NS_Green = NS_GREEN)
NS Yellow	(NS_Yellow = NS_YELLOW)
NS Red	(NS_Red = EW_GREEN + EW_YELLOW)
EW Green	(EW_Green = EW_GREEN)
EW Yellow	(EW_Yellow = EW_YELLOW)
EW Red	(EW_Red = NS_GREEN + NS_YELLOW)

These expressions directly describe the output behavior of the FSM used in the Verilog implementation.

Karnaugh Map (K-Map) Simplification

Karnaugh Maps (K-Maps) are used to simplify Boolean expressions and reduce the number of logic gates required in a digital circuit.

For the Traffic Signal Countdown System, the traffic-light outputs depend on the four FSM states. Since there are four states, **two state variables** are sufficient to represent them. For the K-Map, the states can therefore be represented using two state bits.

The state encoding used for the K-Map is:

State	State Bits
NS_GREEN	0
NS_YELLOW	1
EW_GREEN	10
EW_YELLOW	11

The output truth table is therefore converted into the following K-Map values.

K-Map for NS Green

NS Green is ON only in the NS_GREEN state.

Q1 \Q0	0	1
0	**1**	0
1	0	0

The only 1 occurs when: Q1 =0,Q0 =0
Therefore: NS_Green = Q1 Q0

K-Map for NS Yellow

NS Yellow is ON only in the NS_YELLOW state.

Q1 \Q0	0	1
0	0	**1**
1	0	0

The only 1 occurs when: Q1 =0,Q0 =1
Therefore: NS_Yellow = Q1 Q0

K-Map for NS Red

NS Red is ON during both East-West states.

Q1 \Q0	0	1
0	0	0
1	**1**	**1**

The two 1s can be grouped together. Since Q0 changes between 0 and 1, it is eliminated.
Therefore: NS_Red = Q1

K-Map for EW Green

EW Green is ON only in the EW_GREEN state.

Q1 \Q0	0	1
0	0	0
1	**1**	0

Therefore: EW_Green = Q1 Q0

K-Map for EW Yellow

EW Yellow is ON only in the EW_YELLOW state.

Q1 \Q0	0	1
0	0	0
1	0	**1**

Therefore: EW_Yellow = Q1 Q0

K-Map for EW Red

EW Red is ON during both North-South states.

Q1 \Q0	**0**	**1**
0	**1**	**1**
1	0	0

The two 1s can be grouped together. Q0 is eliminated because it changes between 0 and 1.
Therefore: EW_Red = Q1

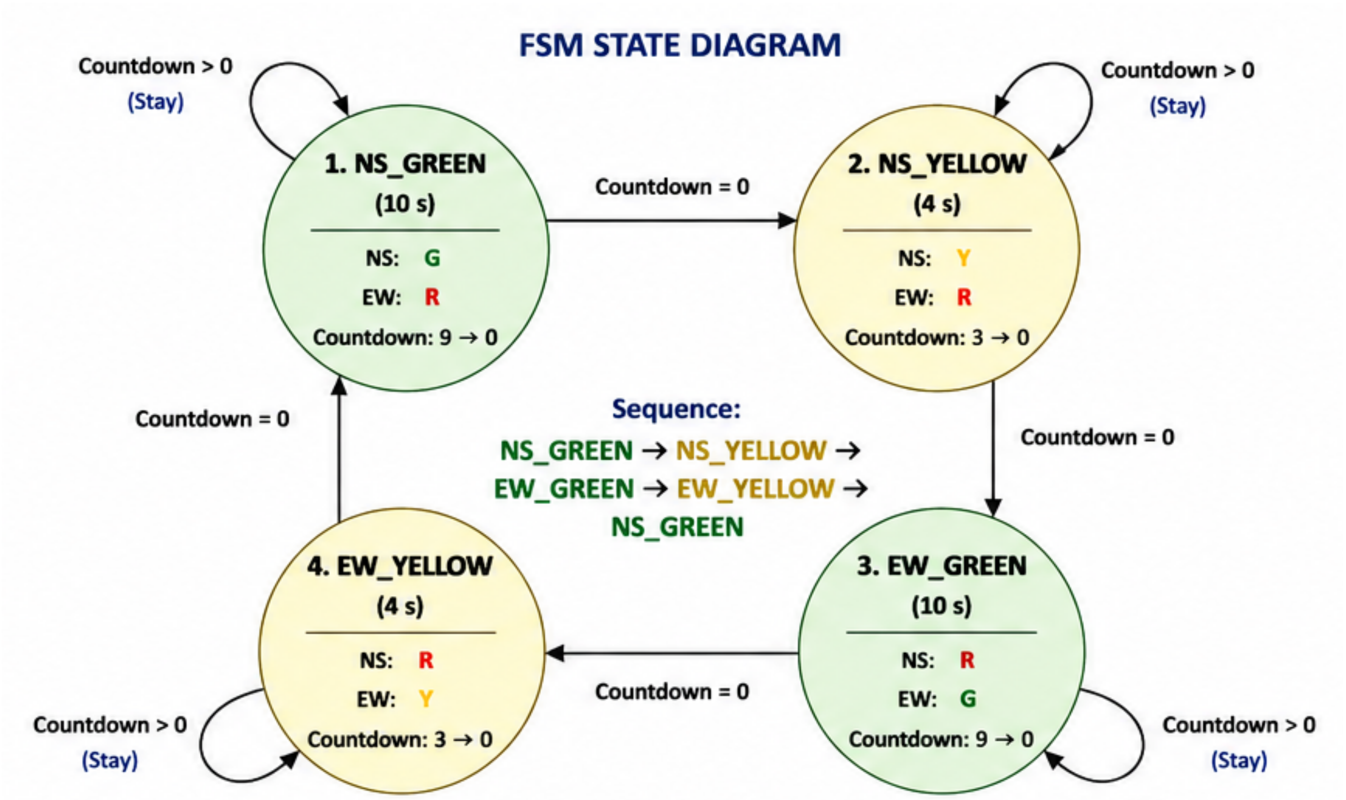
Simplified Boolean Expressions

After K-Map simplification, the expressions are:

Output	Simplified Boolean Expression
NS Green	Q1 Q0
NS Yellow	Q1 Q0
NS Red	Q1
EW Green	Q1 Q0
EW Yellow	Q1 Q0
EW Red	Q1

The K-Map simplification reduces the required logic and makes the traffic-light control circuit simpler. The simplified expressions also clearly show that the two traffic directions are controlled in a complementary manner: when North-South has a Green or Yellow signal, East-West remains Red, and vice versa.

Finite State Machine Design



LEGEND

- G = Green
- Y = Yellow
- R = Red

COUNTDOWN:

- Green State = 9 → 0 (10 s)
- Yellow State = 3 → 0 (4 s)

STATE TIMING

- NS_GREEN = 10 seconds
- NS_YELLOW = 4 seconds
- EW_GREEN = 10 seconds
- EW_YELLOW = 4 seconds

TOTAL CYCLE TIME = 28 seconds

STATE TRANSITION TABLE

Current State	Condition	Next State	Action
NS_GREEN	Countdown > 0	**NS_GREEN**	Stay in NS_GREEN, Countdown = Countdown - 1
NS_GREEN	Countdown = 0	**NS_YELLOW**	Load 3, Countdown = 3
NS_YELLOW	Countdown > 0	**NS_YELLOW**	Stay in NS_YELLOW, Countdown = Countdown - 1
NS_YELLOW	Countdown = 0	**EW_GREEN**	Load 9, Countdown = 9
EW_GREEN	Countdown > 0	**EW_GREEN**	Stay in EW_GREEN, Countdown = Countdown - 1
EW_GREEN	Countdown = 0	**EW_YELLOW**	Load 3, Countdown = 3
EW_YELLOW	Countdown > 0	**EW_YELLOW**	Stay in EW_YELLOW, Countdown = Countdown - 1
EW_YELLOW	Countdown = 0	**NS_GREEN**	Load 9, Countdown = 9

OUTPUTS IN EACH STATE

State	North-South Lights			East-West Lights		
	Red	Yellow	Green	Red	Yellow	Green
NS_GREEN	0	0	1	1	0	0
NS_YELLOW	0	1	0	1	0	0
EW_GREEN	1	0	0	0	0	1
EW_YELLOW	1	0	0	0	1	0

NOTES:

- The FSM changes to the next state only when Countdown = 0.
- During Countdown > 0, the FSM remains in the same state.
- Only one direction has Green or Yellow at a time; the other direction is Red.

Circuit Design

The Traffic Signal Countdown System was designed and tested using two approaches: Verilog simulation in Vivado and a hardware-based circuit simulation in Proteus. Both implementations follow the same basic traffic-light sequence and countdown operation.

Main Components of the System

The complete system consists of the following main blocks:

1. Clock Generator – provides the clock pulses required for the system.
2. Finite State Machine (FSM) – controls the traffic-light sequence.
3. Countdown Counter – counts down the remaining time for the current traffic-light state.
4. Traffic Light Control – generates the Red, Yellow, and Green outputs for both directions.
5. 7-Segment Display – displays the remaining countdown value.
6. Traffic LEDs – represent the Red, Yellow, and Green signals in the Proteus circuit.

Verilog Design

The Verilog implementation contains the main control logic of the system.
The FSM contains four states:

- NS_GREEN
- NS_YELLOW
- EW_GREEN
- EW_YELLOW

The FSM changes state when the countdown reaches zero.

The countdown values used in the design are:

State	Countdown
NS_GREEN	9 → 0
NS_YELLOW	3 → 0
EW_GREEN	9 → 0
EW_YELLOW	3 → 0

The Verilog design also generates the traffic-light outputs according to the current state. For example, during NS_GREEN, the North-South Green output is HIGH while the East-West Red output is HIGH.

The Verilog implementation was simulated in Vivado to verify the state transitions, countdown operation, and traffic-light outputs.

Proteus Circuit Design

The Proteus implementation represents the same traffic-control concept using digital components. The main components used in the Proteus circuit include:

- NE555 timer for generating the clock signal
- 74190 counter for the countdown operation
- 7485 comparator for detecting the required count
- 7447 BCD-to-7-segment decoder for displaying the countdown
- CD4017 decade counter for controlling the traffic-light sequence
- Red, Yellow, and Green LEDs for the traffic signals
- Resistors and capacitors for the required timing and LED connections
- 9 V battery supply

The NE555 generates periodic clock pulses. These pulses are supplied to the counter, causing the displayed value to change with each clock pulse.

The counter loads the required initial value for each traffic-light period and counts down toward zero. When the required count is reached, the control circuit causes the traffic signal to change to the next stage.

The 7447 decoder converts the counter's BCD output into the corresponding signals required to drive the 7-segment display.

The LEDs indicate the current traffic condition:

- Green: traffic is allowed to move.
- Yellow: traffic should prepare to stop.
- Red: traffic must stop.

Overall Circuit Operation

The complete operation can be summarized as follows:

```
Clock Generator
      ↓
Countdown Counter
      ↓
Count reaches 0
      ↓
Change Traffic-Light State
      ↓
Load Next Countdown Value
      ↓
Display Countdown
      ↓
Traffic LEDs Change
      ↓
Repeat
```

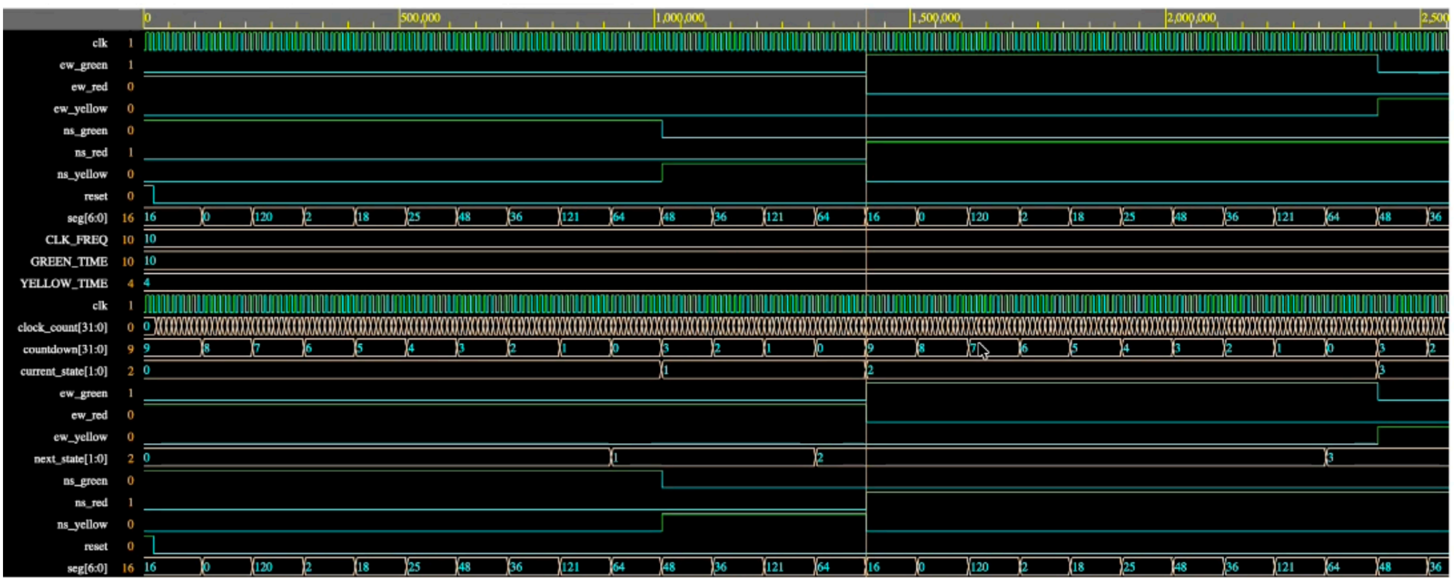
Simulation Results

The Traffic Signal Countdown System was tested using both Vivado and Proteus simulations. The simulations were used to verify the traffic-light sequence, countdown operation, and state transitions.

Vivado Simulation

The Verilog design was simulated in Vivado. The waveform shows the four traffic-light states operating in the required sequence:

NS_GREEN→NS_YELLOW→EW_GREEN→EW_YELLOW
The countdown decreases during each state and, when it reaches zero, the system changes to the next state. The simulation confirms that the traffic-light outputs change according to the FSM design.

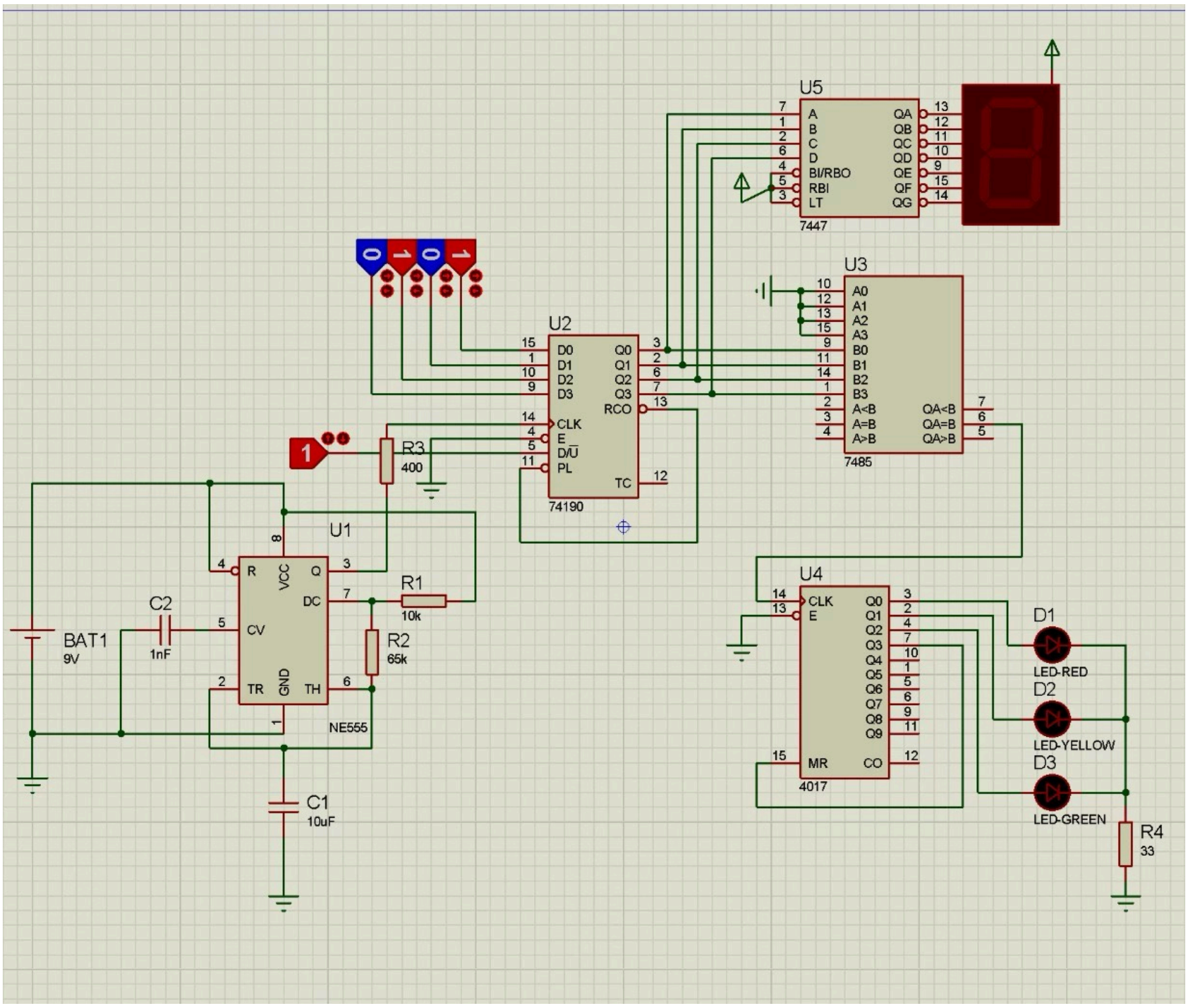


Vivado simulation waveform of the Traffic Signal Countdown System.

Proteus Simulation

The circuit was also simulated in Proteus. The simulation shows the traffic LEDs changing according to the current state, while the 7-segment display shows the remaining countdown value.

The Proteus simulation demonstrates the same basic sequence as the Verilog implementation, with the countdown reaching zero before the traffic signal changes to the next stage.

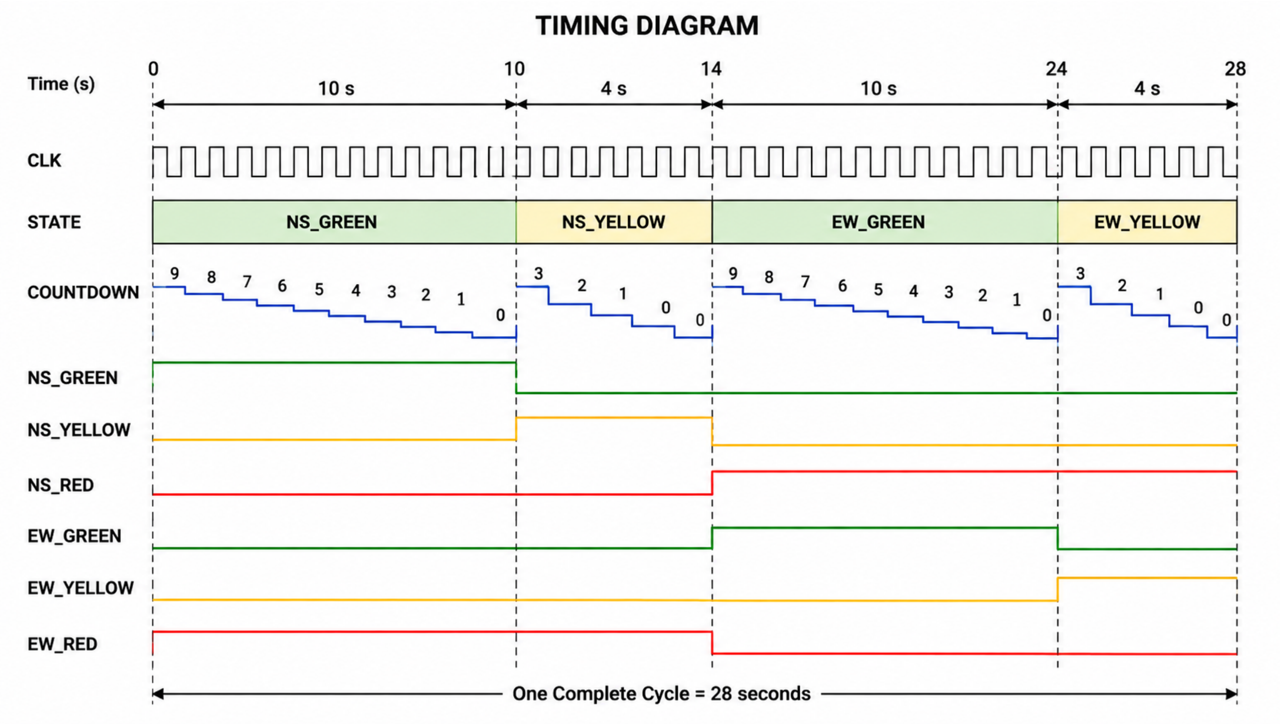


Proteus simulation of the Traffic Signal Countdown System.

Simulation Verification

The results from both simulations show that the system performs the required traffic signal sequence and countdown operation. The Vivado simulation verifies the Verilog/FSM logic, while the Proteus simulation verifies the circuit-level implementation.

Timing Diagram



The timing diagram shows how the traffic-light states and countdown change with respect to the clock signal. The system operates through four states in a fixed sequence:

NS_GREEN→NS_YELLOW→EW_GREEN→EW_YELLOW
The timing sequence is:

State	Countdown	Duration
NS_GREEN	$9 \to 0$	10 seconds
NS_YELLOW	$3 \to 0$	4 seconds
EW_GREEN	$9 \to 0$	10 seconds
EW_YELLOW	$3 \to 0$	4 seconds

During each state, the countdown decreases with every clock pulse. When the countdown reaches zero, the FSM changes to the next state and the appropriate countdown value is loaded.

The complete cycle takes:

$$10+4+10+4=28 \text{ seconds}$$

Challenges and Solutions

During the development of the Traffic Signal Countdown System, several challenges were encountered during the design and simulation process.

Designing the FSM

One of the main challenges was designing the correct sequence of traffic-light states. The system had to move through Green, Yellow, and Red conditions in the correct order without allowing conflicting Green signals.

Solution:

The traffic signals were divided into four clear states: NS_GREEN, NS_YELLOW, EW_GREEN, and EW_YELLOW. The FSM was then designed to move between these states in a fixed sequence.

Countdown Control

Another challenge was making sure that the traffic signal changed only after the countdown reached zero.

Solution:

The countdown was connected to the state-transition logic. The system remains in the current state while the countdown is running and changes to the next state when the countdown reaches zero.

Verifying the Verilog Design

It was necessary to confirm that the Verilog code was producing the expected state transitions and output signals.

Solution:

The design was simulated in Vivado using a testbench. The waveform was observed to verify the countdown, state changes, and traffic-light outputs.

Proteus Circuit Simulation

Another challenge was representing the countdown and traffic-light control using individual digital components in Proteus.

Solution:

The circuit was divided into separate functional parts, including the clock generator, counter, comparator, 7-segment display, and traffic LEDs. This made it easier to test and identify problems in the circuit.

Conclusion and Future Improvements

Conclusion

The Traffic Signal Countdown System was successfully designed and simulated using Verilog in Vivado and Proteus. The system uses a finite state machine to control the traffic-light sequence and a countdown mechanism to determine when the signal should change.

The Vivado simulation verified the digital logic, state transitions, countdown operation, and output signals through the waveform. The Proteus simulation demonstrated the corresponding circuit operation using traffic LEDs and a 7-segment countdown display.

Overall, the project demonstrates how a traffic signal system can be designed using digital logic, FSM concepts, counters, and Verilog HDL.

Future Improvements

The system could be improved in several ways in the future:

- Add pedestrian crossing signals and pedestrian push buttons.
- Add vehicle sensors so that signal timing can change according to traffic conditions.
- Replace the fixed countdown values with programmable timing.
- Add a larger or clearer display for the countdown.
- Implement the Verilog design on an FPGA development board for physical hardware testing.
- Add an emergency vehicle priority system.
- Make the traffic-light timing adaptive instead of using fixed time periods.